Development of a Compartmentalized Metabolic Subnetwork for the Endothelial Cytoskeleton to Model SHock-Induced Endotheliopathy (SHINE)

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1. Introduction: The Biological and Modeling Problem

1.1. Pathophysiology of SHock-Induced Endotheliopathy (SHINE)

SHock-Induced Endotheliopathy (SHINE) is a critical condition where systemic shock leads to widespread injury of the endothelial cells lining blood vessels. This "endotheliopathy" is characterized by the breakdown of the endothelial barrier, leading to vascular leakage, edema, and subsequent multiple organ failure. The molecular drivers of SHINE include a massive inflammatory response and sympatho-adrenal hyperactivation.

1.2. The Cytoskeleton as a Central Player in Endothelial Dysfunction

The integrity of the endothelial barrier is physically maintained by the cell's cytoskeleton—a dynamic network of protein filaments. In SHINE, endothelial cells undergo a dramatic pathological reorganization of this network, shifting from a stable state to a hyper-contractile state characterized by the formation of actin stress fibers. This process actively pulls cells apart, creating intercellular gaps and driving vascular leakage.

1.3. The Modeling Objective: Transcending the Limitations of Static Representations

The main goal was to develop a metabolic model capable of capturing the unique bioenergetic demands of the endothelial cytoskeleton during SHINE. Initial approaches in genome-scale models (GEMs) often use "demand" reactions to represent the cost of building cellular components. However, the pathology of SHINE is not simply the synthesis of *more* cytoskeleton, but the frantic, energy-intensive *remodeling* of the existing structure. This required a move from a static model of composition to a dynamic model of function.

2. Phase 1: The Initial Approach – A Single Aggregated Demand Reaction

2.1. Theoretical Basis: The Role of Demand Reactions in GEMs

The first logical step was to create a single demand reaction. In a GEM, this is a sink reaction that consumes a stoichiometrically defined set of precursors (amino acids, ATP, etc.) to represent the production of a specific macromolecule or the entire cell biomass.

2.2. Quantitative Composition of the Structural Cytoskeleton

To build this reaction, we first defined the bulk composition of the primary structural filaments of the cytoskeleton.

Table 2.1: Initial 3-Component Mass Distribution

Component	Mass Percentage	Primary Protein(s)	UniProt ID(s) Used
Actin Filaments	50%	Actin, cytoplasmic 1	P60709

Microtubules	30%	Tubulin alpha-1A & Tubulin beta	P68363, P07437
Intermediate Filaments	20%	Vimentin	P08670

2.3. Derivation and Formulation of the Initial Demand Reaction

Based on the weighted average of the amino acid compositions from the proteins listed in Table 2.1, and including the energy cost for polymerization (ATP for actin, GTP for tubulin), the following single demand reaction was formulated to represent the synthesis of 1 gram of cytoskeleton.

Table 2.2: Detailed Stoichiometry of the Aggregated Reaction

Metabolite	mmol required per 1g Cytoskeleton
L-Alanine	0.776
L-Arginine	0.435
L-Asparagine	0.389
L-Aspartate	0.814
L-Cysteine	0.222
L-Glutamine	0.403
L-Glutamate	1.056
Glycine	0.867
L-Histidine	0.224
L-Isoleucine	0.540
L-Leucine	0.825
L-Lysine	0.505
L-Methionine	0.218
L-Phenylalanine	0.380
L-Proline	0.463
L-Serine	0.627
L-Threonine	0.518
L-Tryptophan	0.106
L-Tyrosine	0.301
L-Valine	0.655
ATP (+ H_2O)	5.86
GTP (+ H_2O)	2.94

3. Phase 2: The Critical Turning Point – Identifying the Model's Limitations

3.1. The Fundamental Flaw: Conflating Static Synthesis with Dynamic Remodeling

A critical analysis revealed that the single demand reaction was fundamentally unsuitable for modeling SHINE. It rigidly links the energy cost (ATP/GTP) to the synthesis of new protein mass (amino acids). However, the pathophysiology of SHINE is dominated by a massive increase in energy consumption for the *reorganization* of the *existing* protein pool (e.g., actin treadmilling, actomyosin contraction). This remodeling process consumes energy at a rate independent of, and far greater than, new protein synthesis.

3.2. The Pathophysiological Imperative: Decoupling Energy Costs for SHINE

The key insight was the need to decouple these two distinct metabolic burdens. A valid model must be able to independently simulate:

1. **The Cost of Synthesis:** The drain of amino acids and a baseline ATP cost for creating new proteins to replace damaged ones.
2. **The Cost of Remodeling:** A separate, variable, and much larger energy drain (ATP/GTP hydrolysis) representing the hyper-dynamic state of the cytoskeleton during SHINE.

3.3. Conclusion: Abandoning the Single Reaction for a Subnetwork Approach

The single demand reaction was discarded in favor of a more sophisticated framework that could treat the cytoskeleton as a dynamic, energy-consuming subsystem.

4. Phase 3: Design of the Advanced Subnetwork

4.1. Conceptual Framework: A Virtual Cytoskeleton Compartment ([ck])

The chosen solution was to create a new, virtual "cytoskeleton" compartment ([ck]) within the model. This design provides modularity, allows for the representation of local metabolite pools, and is extensible for future additions.

4.2. Refining the Framework: The "Local Synthesis" Model

The most advanced model design, and the one we adopted, was the **"local synthesis"** model. Instead of synthesizing proteins in the cytosol ([c]) and transporting the finished products, this model transports the basic building blocks (amino acids, ATP) into the [ck] compartment and simulates protein synthesis *in situ*. This reflects biological evidence for localized translation of cytoskeletal components and offers the most granular control.

4.3. Expanding Biochemical Scope: Incorporating Regulatory Glycans and Lipids

To increase the model's biological realism, key regulatory molecules with direct metabolic links were incorporated:

- **Glycans (O-GlcNAcylation):** The substrate **UDP-GlcNAc**, derived from glucose, was added to model the post-translational modification of proteins like vimentin.
- **Lipids (PIP2): Phosphatidylinositol 4,5-bisphosphate (PIP2)** was included as a key regulator of actin dynamics at the plasma membrane.

5. Phase 4: Curation, Refinement, and Data Integration

5.1. Analysis of External Metabolic Data

A list of potential metabolites and reactions for the [ck] compartment was analyzed to identify which processes were truly localized to the cytoskeleton versus being part of general cytosolic metabolism.

Table 5.1: Analysis of Metabolites from compartment_ck_metabolites.csv
Metabolite	ID	Include in [ck]?	Justification

ATP, ADP, GTP, GDP, Pi, H+, H2O	Various	**Yes**	Direct substrates/products for polymerization and contraction.
All Amino Acids	Various	**Yes**	Substrates for local synthesis.
[myosin light chain]	MAM00190ck	**Yes**	Direct substrate for the contractile machinery.
PIP2 Precursors	MAM00553ck	**Yes**	Local synthesis of the lipid regulator PIP2.
Nitric oxide	MAM02609ck	No	Diffusible signaling molecule produced in the cytosol.
Glucose, Succinate, Zymosterol	Various	No	Part of central metabolism in other compartments (cytosol, mitochondria).

Table 5.2: Analysis of Reactions from compartment_ck_reactions.csv

Reaction	ID	Include in [ck]?	Justification
myosin light chain + ATP --> ...	MAR22907	**Yes**	The core reaction of actomyosin contraction.
ADP + GTP <=> ATP + GDP	MAR13466	**Yes**	Local regeneration of GTP for tubulin polymerization.
ATP + PI(4)P --> PIP2 + ADP	MAR16740	**Yes**	Local synthesis of the regulatory lipid PIP2.
L-Arginine + ... --> Nitric oxide	MAR23221	No	Nitric Oxide Synthase (NOS) is a cytosolic enzyme.
Branched-chain aa catabolism	MAR18983	No	This is a mitochondrial pathway.
Fatty acid synthesis	MAR14567	No	This is a cytosolic pathway.

5.2. Iterative Debugging: Identifying and Resolving Blocked Reactions

The initial subnetwork design was found to contain blocked reactions because several byproducts were "dead-end" metabolites with no exit pathway.

Table 5.3: Identified Dead-End Metabolites and Blocked Reactions

Dead-End Metabolite	Produced by Reactions...
pi[ck] (Phosphate)	Actin_Remodeling, Tubulin_Remodeling, Myosin_Dephosphorylation
h[ck] (Proton)	Actin_Remodeling, Tubulin_Remodeling
udp[ck] (Uridine diphosphate)	Vimentin_Glycosylation, Actin_Modification

To resolve this, reversible export transport reactions were added.

Table 5.4: Corrective Export Reactions to Unblock the Network

Reaction ID	Reaction Equation
T_Pi_out	pi[ck] <=> pi[c]
T_H_out	h[ck] <=> h[c]
T_UDP_out	udp[ck] <=> udp[c]

5.3. Final Integration: The Myosin Contractile Machinery

A final curation step identified that the myosin light chain was being treated as an external input, inconsistent with the local synthesis model. To correct this, the transport of the pre-made protein was removed, and a new local synthesis reaction was added.

Table 5.5: Data for Myosin Light Chain 9 Synthesis

Protein	Myosin light polypeptide 9 (MYL9)

UniProt ID	P24844
Length	172 amino acids
Translation Cost	684 ATP

6. Phase 5: Finalizing the Model Composition

6.1. Revisiting the Mass Distribution: Incorporating Myosin

With the integration of myosin, the initial 3-component composition was re-evaluated to create a more comprehensive 4-component model that includes the contractile machinery. Myosin II was estimated to contribute ~4% of the total mass of the core cytoskeletal system.

6.2. The Final Reference Composition for Flux Analysis

This revised composition serves as the definitive baseline for setting realistic fluxes in the final, flexible subnetwork model.

Table 6.1: Revised 4-Component Cytoskeletal Mass Distribution

Component	Revised Percentage	Role
Actin Filaments	48.0%	Structural Polymer (Scaffold)
Microtubules (Tubulin)	28.8%	Structural Polymer (Tracks)
Intermediate Filaments (Vimentin)	19.2%	Structural Polymer (Mechanical Strength)
Myosin II Motor Protein	4.0%	Motor Protein (Force Generation)
Total	100.0%	

7. Final Model Specification: The Complete Reaction Set

This is the complete, final list of all reactions in the cytoskeleton subnetwork.

7.1. Block 1: Substrate Transport (from [c] to [ck])

26 reversible reactions

Reaction ID	Reaction Equation
T_Ala to **T_Val**	(amino_acid)[c] <=> (amino_acid)[ck] (20 reactions)
T_ATP	atp[c] <=> atp[ck]
T_GTP	gtp[c] <=> gtp[ck]
T_UDP_GlcNAc	udp_glcnac[c] <=> udp_glcnac[ck]
T_PI4P	pi4p[c] <=> pi4p[ck]
T_H2O	h2o[c] <=> h2o[ck]

7.2. Block 2: Macromolecule Synthesis (in Cytoskeleton [ck])

4 irreversible reactions

Reaction ID	Reaction Equation
R_Synth_Actin	22 ala-L[ck] + 18 arg-L[ck] + 15 asn-L[ck] + 30 asp-L[ck] + 16 cys-L[ck] + 24 gln-L[ck] + 26 glu-L[ck] + 28 gly[ck] + 8 his-L[ck] + 29 ile-L[ck] + 26 leu-L[ck] + 20 lys-L[ck] + 16 met-L[ck] + 15 phe-L[ck] + 17 pro-L[ck] + 21 ser-L[ck] + 27 thr-L[ck] + 4 trp-L[ck] + 15 tyr-

L[ck] + 18 val-L[ck] + 1496 atp[ck] + 1122 h2o[ck] --> actin_protein_bare[ck] + 1496 adp[ck] + 1496 pi[ck] + 1496 h[ck] |
R_Synth_Tubulin	61 ala-L[ck] + 40 arg-L[ck] + 37 asn-L[ck] + 62 asp-L[ck] + 23 cys-L[ck] + 35 gln-L[ck] + 83 glu-L[ck] + 74 gly[ck] + 22 his-L[ck] + 46 ile-L[ck] + 71 leu-L[ck] + 42 lys-L[ck] + 22 met-L[ck] + 42 phe-L[ck] + 46 pro-L[ck] + 62 ser-L[ck] + 55 thr-L[ck] + 12 trp-L[ck] + 36 tyr-L[ck] + 65 val-L[ck] + 3580 atp[ck] + 2685 h2o[ck] --> tubulin_dimer_bare[ck] + 3580 adp[ck] + 3580 pi[ck] + 3580 h[ck]
R_Synth_Vimentin	36 ala-L[ck] + 25 arg-L[ck] + 21 asn-L[ck] + 37 asp-L[ck] + 6 cys-L[ck] + 28 gln-L[ck] + 53 glu-L[ck] + 20 gly[ck] + 7 his-L[ck] + 20 ile-L[ck] + 44 leu-L[ck] + 23 lys-L[ck] + 8 met-L[ck] + 13 phe-L[ck] + 17 pro-L[ck] + 37 ser-L[ck] + 26 thr-L[ck] + 3 trp-L[ck] + 16 tyr-L[ck] + 26 val-L[ck] + 1860 atp[ck] + 1395 h2o[ck] --> vimentin_protein_bare[ck] + 1860 adp[ck] + 1860 pi[ck] + 1860 h[ck]
R_Synth_MyosinLC	15 ala-L[ck] + 5 arg-L[ck] + 9 asn-L[ck] + 14 asp-L[ck] + 1 cys-L[ck] + 6 gln-L[ck] + 19 glu-L[ck] + 10 gly[ck] + 2 his-L[ck] + 8 ile-L[ck] + 10 leu-L[ck] + 14 lys-L[ck] + 6 met-L[ck] + 8 phe-L[ck] + 6 pro-L[ck] + 7 ser-L[ck] + 8 thr-L[ck] + 0 trp-L[ck] + 3 tyr-L[ck] + 8 val-L[ck] + 684 atp[ck] + 513 h2o[ck] --> myosin_light_chain[ck] + 684 adp[ck] + 684 pi[ck] + 684 h[ck]

7.3. Block 3: Local Regulation & Protein Modification (in Cytoskeleton [ck])

4 reactions

Reaction ID	Reaction Equation
R_PIP2_Synth	pi4p[ck] + atp[ck] --> pip2[ck] + adp[ck] + h[ck]
R_GTP_Regen_NDPK	gdp[ck] + atp[ck] <=> gtp[ck] + adp[ck]
R_Vim_Glyco	vimentin_protein_bare[ck] + udp_glcnac[ck] --> vimentin_GlcNAc[ck] + udp[ck]
R_Actin_Mod	actin_protein_bare[ck] + udp_glcnac[ck] + pip2[ck] --> actin_complex[ck] + udp[ck]

7.4. Block 4: Energy Dissipation & Cytoskeletal Function (in Cytoskeleton [ck])

3 irreversible reactions

Reaction ID	Reaction Equation
R_Myosin_Contract	myosin_light_chain[ck] + atp[ck] --> myosin_light_chain_P[ck] + adp[ck]
R_Actin_Remodel	atp[ck] + h2o[ck] --> adp[ck] + pi[ck] + h[ck]
R_Tubulin_Remodel	gtp[ck] + h2o[ck] --> gdp[ck] + pi[ck] + h[ck]

7.5. Block 5: Product Turnover & Waste Export (in and out of [ck])

10 reactions

Reaction ID	Reaction Equation
R_Myosin_Dephospho	myosin_light_chain_P[ck] + h2o[ck] --> myosin_light_chain[ck] + pi[ck]
DM_Actin	actin_complex[ck] -->

DM_Tubulin	tubulin_dimer_bare[ck] -->
DM_Vimentin	vimentin_GlcNAc[ck] -->
DM_MyosinLC	myosin_light_chain[ck] -->
T_ADP_out	adp[ck] <=> adp[c]
T_GDP_out	gdp[ck] <=> gdp[c]
T_Pi_out	pi[ck] <=> pi[c]
T_H_out	h[ck] <=> h[c]
T_UDP_out	udp[ck] <=> udp[c]

8. Conclusion: A Mechanistically Detailed and Robust Modeling Tool

The process documented in this report has yielded a compartmentalized metabolic subnetwork of the endothelial cytoskeleton. By progressing from a simple demand reaction to this final, integrated model, we have created a tool that is mechanistically suited for studying the complex pathophysiology of SHINE. The final model correctly decouples the metabolic costs of protein synthesis from the massive energy demands of dynamic remodeling and contraction, incorporates key regulatory inputs from glucose and lipid metabolism, and is structurally unblocked. This subnetwork now serves as a robust platform for generating testable hypotheses about the metabolic vulnerabilities of endothelial cells cytoskeleton during critical illness.

9. References and Data Sources

1. **UniProt Consortium.** (2023). UniProt: the universal protein knowledgebase in 2023. *Nucleic Acids Research*, 51(D1), D523–D531.
    - **Relevance:** Authoritative source for the primary amino acid sequences used to determine the exact stoichiometric requirements for the synthesis reactions of Actin (P60709), Tubulin alpha-1A (P68363), Tubulin beta (P07437), Vimentin (P08670), and Myosin Light Chain 9 (P24844).
2. **Thiele, I., & Palsson, B. Ø.** (2010). A protocol for generating a high-quality genome-scale metabolic reconstruction. *Nature Protocols*, 5(1), 93–121.
    - **Relevance:** Provided the standard methodology for constructing macromolecule synthesis reactions from primary sequence data and for accounting for energy costs, which guided the entire model-building process.
3. **Johansson, P. I., Stensballe, J., & Ostrowski, S. R.** (2017). Shock induced endotheliopathy (SHINE) in acute critical illness - a unifying pathogenic mechanism. *Critical Care*, 21(1), 25.
    - **Relevance:** Formally defined SHINE, providing the essential biological and clinical context for our modeling objective.
4. **Ince, C., et al.** (2016). The endothelium in sepsis. *Shock*, 45(3), 259–270.
    - **Relevance:** Justified the focus on *remodeling* of the actin cytoskeleton (stress fiber formation) as the key pathological and energy-consuming event in endothelial dysfunction.
5. **Dudek, S. M., & Garcia, J. G. N.** (2001). Cytoskeletal regulation of pulmonary vascular permeability. *Journal of Applied Physiology*, 91(4), 1487–1500.

- **Relevance:** Provided the specific biochemical justification for creating high-flux energy dissipation reactions, detailing the ATP-dependent nature of actomyosin contraction.

6. **Hart, G. W., et al.** (2011). Cross talk between O-GlcNAcylation and phosphorylation: roles in signaling, transcription, and disease. *Annual Review of Biochemistry*, 80, 825–858.
   - **Relevance:** Justified the inclusion of O-GlcNAcylation by identifying cytoskeletal proteins as key targets of this nutrient-sensing modification, linking glucose metabolism to cytoskeletal function.
7. **Yin, H. L., & Janmey, P. A.** (2003). Phosphoinositide regulation of the actin cytoskeleton. *Annual review of physiology*, 65, 761-789.
   - **Relevance:** Provided the rationale for including lipid metabolism by detailing the critical role of PIP2 in regulating actin-associated proteins at the cell membrane.